



ParcelResolve

Quality Assurance Testing

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SUMMARY

Overview

ParcelResolve is a WhatsApp-based assistant that uses AI to automate the identification and reporting of damaged parcel shipments.

Key features and capabilities

The system features a direct Twilio integration for mobile reporting, automated tracking number extraction via Google Gemini, and standardized damage grading into four specific levels. It also includes built-in input validation to ensure photos are attached before generating structured JSON reports.

Benefits

By replacing manual data entry with AI vision, the tool significantly increases reporting speed and removes human bias from damage assessments. It ensures high data integrity by delivering machine-readable results that integrate directly into existing logistics databases.

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PROBLEM

- Manual transcription of tracking codes is slow and prone to frequent data entry errors.
- Current damage assessments are highly subjective and lack a standardized grading protocol.

SOLUTION

- The system utilizes a WhatsApp-native interface to turn any smartphone into an automated reporting tool.
- Multimodal AI standardizes damage levels and extracts tracking data into structured JSON formats.

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RESULTS

- **High-Fidelity Data Extraction:** The system successfully demonstrated its ability to accurately "read" physical parcel labels and pull specific tracking numbers, such as **MY999888777** and **MY123456789**, directly from user-submitted WhatsApp photos. This confirms that the multimodal AI pipeline can reliably isolate alphanumeric codes from complex, real-world backgrounds without any manual data entry required from the user.
- **Consistent Damage Grading:** Across multiple test cases ranging from perfectly intact boxes to severely crushed packages, the bot consistently assigned the correct standardized damage level—categorizing them as **NONE, LOW, or HIGH**. These results prove that the system can remove human bias from the reporting process, providing the logistics team with a uniform and objective assessment of every parcel's condition.
- **End-to-End Workflow Automation:** Beyond simple data analysis, the results show that the platform can successfully trigger secondary business logic, such as automatically initiating a replacement for parcels flagged with a **"HIGH"** damage level. This evidence proves that the system isn't just a chatbot, but a functional logistics tool that can hand off critical data to specialized agents to resolve customer issues in real-time.



Tracking MY99888777 is broken 04:08 ✓

Processed by Agent A:
{ "trackingNumber": "MY99888777", "damageLevel": "LOW" } 04:08



Tracking MY123456789 is broken 04:26 ✓

Processed by Agent A:
{ "trackingNumber": "MY123456789", "damageLevel": "NONE" } 04:26

Processed by Agent A:
{`"trackingNumber": "MY987654321", "damageLevel": "HIGH"`} 04:29



Help!! my parcel is broken ,Tracking
MY987654321

04:28 ✓✓

Agent C has reviewed your case. We are processing a replacement for your damaged parcel. 04:47



Help!! my parcel is broken ,Tracking
MY987654321

04:47 ✓✓